

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in [linkedin.com/in/muhammad-haider-zaidi](https://www.linkedin.com/in/muhammad-haider-zaidi)
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, C++, Javascript

Deep Learning & ML: Machine Learning, Predictive Modeling, Statistical Modeling, Deep Learning, Computer Vision, NLP, LLMs

Data Tools: SQL, Python (Pandas, NumPy, Scikit-learn), Data Visualization (Tableau, PowerBI, Matplotlib, Seaborn)

MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, Real-time System Optimization

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Data Ingestion

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **predictive models** and **deep learning solutions** for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability, contributing to advanced **statistical modeling**.
- **Collaborated closely with cross-functional** PhD researchers on publications; created **data visualizations** for **model performance monitoring** and generating **business insights**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Developed and deployed **machine learning solutions** (AI algorithms) that improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x, demonstrating strong **problem-solving skills**.
- Engineered **predictive models** for a real-time tennis foul detection system using TensorRT, significantly reducing compute cost and time by 6x, enhancing **analytical workflows**.
- Managed MLOps pipelines on AWS SageMaker, addressing model data drift, to ensure robust **machine learning solutions** and improve **data processing workflows**.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Contributed to full-stack development of an AI-based services website, gaining experience in deploying **machine learning solutions** within web environments.
- Developed an AI chatbot application utilizing OpenAI API and LLM frameworks, focusing on natural language processing to generate **business insights** and support user interaction.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built **web scrapers** and **ETL pipelines** to process **large datasets**, automating **data ingestion and cleaning workflows** for clients to generate accurate **business insights**.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected and enhanced **predictive model** accuracy by 4% through a weighted-probability ensemble and custom feature engineering, demonstrating advanced **statistical modeling** and **analytical problem-solving skills**.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time **predictive model** for Bitcoin price forecasting using time-series analysis and **machine learning solutions**; utilized **Python** and **SQL** to process **large datasets** and generate **business insights**.

Saaz: AI Self-Playing Chitrali Sitar

2023

Signal Processing, Music Generation

- Designed a real-time signal processing system for music generation, demonstrating expertise in complex **data processing and analytical workflows**.
- Implemented transformer-based **predictive models** for sequence prediction in audio generation, showcasing advanced **machine learning** capabilities.

Hugging Face

Open-Source

Contributor

- Co-authored technical documentation and implementation for Zero-shot Image Classification on Hugging Face Hub, demonstrating understanding of **machine learning solutions**.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python